

their performance of dozens of algorithms, for catching, throwing, kicking, running, and moving. There are probably also much more complex algorithms, which have not been fully delineated, that govern the operation of the social world of recess: the talking, status seeking, gossiping, alliance formation, and friendship cultivation. When we end recess time and start math class, we take students out of a world of algorithm exploration and push them to learn an unnatural and mechanistic process of solving for x , a process that is not a natural part of human development and is not even the most powerful method for solving analytical problems. Only if students progress to advanced math and computer science do they return to the natural world of algorithms and the powerful processes that they were unconsciously and joyfully mastering at recess.

This book is meant to be an intellectual recess for the curious—a recess in the sense that a young student means it: the beginning of all important activity, the end of all drudgery, and the continuation of cheerful exploration with friends. If you have any feeling of trepidation about algorithms, remind yourself that we humans are naturally algorithmic, and if you can catch a ball or bake a cake, you can master an algorithm.

In the remainder of this book, we explore many different algorithms. Some will sort lists or calculate numbers. Others will enable natural language processing and artificial intelligence. I encourage you to bear in mind that algorithms don't grow on trees. Each algorithm, before it became mainstream and was packaged for general consumption in this book, was discovered or created by someone like Chapman, who woke up one day in a world in which his algorithm didn't exist and went to sleep at the end of that day in a world in which it did. I encourage you to try to get in the mindset of these heroic discoverers. That is, I encourage you to approach an algorithm not only as a tool to be used but also as a formidable problem that was solved. The world of algorithms is not yet close to being fully mapped—many remain to be discovered and perfected, and I earnestly hope that you can be a part of that discovery process.

Summary

In this chapter, you saw two approaches to solving a problem: the analytic one and the algorithmic one. By solving the outfield problem two ways, we